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Cloud computing adoption and its impact on SMEs' performance for cloud supported operations: A dual-stage analytical approach

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ABSTRACT

This paper investigates the key predictors of cloud computing adoption, and further, assesses how cloud computing adoption affects small and medium enterprises' (SMEs') performance. To test the proposed model, we have applied a dual-stage analytical approach by combining structural equation modeling (SEM) and artificial neural network (ANN). SEM results reveal that relative advantage, service quality, perceived risks, top management supports, facilitating conditions, cloud providers influence, server location, computer self-efficacy, and resistance to change have a significant effect on the adoption of cloud computing. Also, this study confirms the positive impact of cloud computing adoption on firm performance. The results of importance-performance map analysis (IPMA) suggest that managerial actions should focus more on improving perceived risk, relative advantage, and top management support. Besides, the results of neural network analysis indicate that the most significant predictor of cloud adoption is server location followed by facilitating conditions, relative advantage, service quality, top management support, computer self-efficacy, perceived risks, cloud provider's influence, and resistance to change. We also discuss the implications of the research that can assist researchers, owners/managers, policymakers, and cloud providers by offering valuable insights regarding cloud computing adoption in SMEs.

1. Introduction

Tremendous market competition and dramatically changing business environment have forced enterprises to adopt various state-of-the-art information technologies (IT) such as Social Networks, Semantic Web, embedded systems, Internet of things (IoT), virtualization technologies, and cloud computing [1,2]. Cloud computing has received more attention in both private and public sectors due to the growing reputation of the shared networks connecting people from all over the world. It offers an innovative business model for the pervasive, convenient, and on-demand access to the virtualized and distributed resources such as network, servers, storage, applications, and services [3]. Cloud services include Software as a Service (SaaS), which offers software applications as services; Platform as a Service (PaaS), which delivers the tools to create and host web application; Infrastructure as a Service (IaaS), which delivers storage and computing; and Expert as a Service (EaaS), which provides human resources (HRs) [4].

Cloud computing increases the capacity of an enterprise since it delivers services on a pay-per-use basis, which allows firms to regulate the usage level of information technology (IT) resources as per current demands. In the age of the Industry 4.0, which has brought digitization and automation in production and operation, the adoption and use of cloud computing have transformed existing enterprise system, delivering enterprises with more agility, flexibility, and productivity [5]. Arvanitis, Kyriakou [6] further stated that cloud computing helps firms to transform the conventional business model, boost effective collaboration, and enhance IT capabilities. Due to its huge potential benefits, firms are rushing to adopt and use cloud computing in their processes and operations. The unique feature (i.e., on-demand service) of cloud computing has made it worthier for small and medium enterprises (SMEs). SMEs have a relatively small IT budget than large firms, which does not allow them to develop IT resources (e.g., hardware, software, IT infrastructure) internally. Hence, SMEs with inadequate technological competence and limited IT experts frequently depend on external IT professionals to achieve its expected outcomes. Cloud computing can create a level-playing-field for them by creating opportunities to utilize

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state-of-art technologies on a rental basis, which previously were unaffordable [7]. As cloud technologies transform the world we live in, better decision-making tools for SMEs will impact their success where SMEs can effectively compete with large businesses leveraging these services [8].

However, cloud computing has not experienced extensive adoption among SMEs [9]. The possible reasons may include lack of industry-specific standards, the paucity of technology readiness, unawareness about potential benefits of this technology, unsatisfactory quality of cloud services, and the incompatibility between initial expectations and real experience. Alshamaila, Papagiannidis [10] identified that lack of awareness of the relative advantage was a major reason for not adopting cloud computing by SMEs. Another critical issue inhibiting the acceptance of cloud computing or leading to their low use is the concern about the security and privacy of resources stored in a third party's server [11,12]. Moreover, decision actors' personal attributes (e.g., self-efficacy, resistance to change, hedonic drives, social influence) may hinder the adoption of cloud computing in SMEs [13,14]. Additionally, supports from cloud providers to organizations are still inadequate to facilitate cloud adoption [10].

The adoption, use, and performance of cloud computing have appeared as an essential research agenda in the scholar and professional environment, particularly for SMEs with inadequate resources. We found several studies in the literature examining factors influencing the adoption of cloud computing in SMEs (see Table 1). However, they overlooked to assess the impact of several important contextual constructs (e.g., server location, risks, cloud provider's influence) on the adoption intention of cloud computing. Essentially, the effect of decision-makers' individual dimension on cloud computing adoption was not sufficiently addressed in the literature. Since SMEs are owned and run by single or a small number of people, strategic and tactical focus, including technology adoption decision, primarily depend on the idiosyncratic given by a small number of decision-makers. Therefore, individual characteristics of decision-makers might have a significant impact on the adoption of cloud computing in the context of SMEs [15].

Besides, extant researchers applied various conventional statistical methods such as descriptive statistics, confirmatory factor analysis (CFA), multiple regression analysis (MRA), structural equation modeling (SEM) for the analysis of data. These techniques only can analyze the linear relationship among factors, which may lead to over-simplification of complicated technology adoption process [16]. Furthermore, prior studies did not examine the relative importance of the predictors of cloud computing adoption. The relative importance helps organizations to concentrate on the most important factor while allocating their resources through the adoption process [17]. Moreover, despite the confirmation of the positive relationship between information systems (IS) adoption and firm performance in existing literature, the assessment of the effect of cloud adoption on the performance had received negligible attention [16,18].

Thus, this research aims to bridge the gap in conventional information systems (IS) literature. This study attempts to answer the following research questions (i) What are the significant factors of cloud computing adoption in SMEs? (ii) What is the relationship between cloud computing adoption and firm performance? (iii) What are the critical factors that need special managerial attention to improve cloud computing adoption and firm performance? and (iv) How much important the factors are to predict cloud computing adoption? We applied a dual-stage analytical approach by integrating Structural Equation Modelling (SEM) and neural network model suggested by many scholars [5,16,19–21] to evaluate the research questions. The strength in path analysis of SEM was applied to investigate the first two questions, while importance-performance map analysis (IPMA) was employed to examine the third question. Further, we applied the artificial neural network (ANN) analysis to assess the last question. This systematic mixed approach provides a comprehensive understanding of the studied topic; at the same time, the advantages of one method can compensate for the disadvantages of another method. The study assists SMEs and cloud providers in recognizing various important adoption factors, which leads to the successful implementation of cloud computing.

Table 1
Literature on cloud computing adoption in SMEs.

Studies	Main model	Statistical techniques	Key factors
(Ooi et al., 2018)	UTAUT and TOE	SEM and NN	Performance expectation, firm size, and absorptive capacity
(Kumar et al., 2017)	–	FA and MRA	Perceived benefits, top management support, competitive pressure, and perceived concerns
(Hassan et al., 2017)	TOE	SEM	IT resources and external pressure
(Doherty et al., 2015)	–	Descriptive	Cost-benefit and continuous availability of service providers
(Gupta et al., 2013)	–	EFA and CFA	Ease of use and convenience, security, and privacy and cost reduction
(Agostino et al., 2013)	–	Descriptive analysis	Software's functionalities level, ubiquitous access, handling big data and implementation cost
(Alshamaila et al., 2013)	TOE	Qualitative analysis	Relative advantage, uncertainty, compatibility, trialability, size, top management support, industry, market scope, supplier efforts, and external computing support
(Makena 2013)	TOE	Qualitative analysis	Top management support, size, technology readiness, relative advantage, competitiveness, complexity, and compatibility

2. Literature review

2.1. Cloud computing adoption in SMEs

Prior literature suggests that the adoption of an IS/IT varies according to organizational size; the impact of various factors on IT/IS adoption might be different between large firms and SMEs. Nevertheless, most of the literature [22–25] highlight large organizations without paying adequate attention to SMEs while studying the adoption behaviour of cloud computing. To this point, we have identified a few studies that examine the determinants of cloud computing adoption and assess the effect of this adoption on SMEs performance. The findings of these studies are summarized in Table 1. These studies were criticized for not including a comprehensive model or for applying traditional statistical analysis. Ooi, Lee [5] performed an empirical study on cloud computing measuring the impacts of various UTAUT factors and organizational factors on firm performance. They found that performance expectancy, firm size, and absorptive capacity had a significant positive effect on innovation, while innovation positively affected firm performance. This study used a sample of 188 manufacturing firms located in Malaysia. As this study focused only manufacturing firms, the findings of this study might not be effective for other industries such as service industry. Similarly, Kumar, Samalia [9] conducted a study on cloud computing adoption by SMEs in the Indian context and stated that perceived benefits, top management support, competitive pressure, and perceived concerns had a significant impact on cloud technology adoption. This study is criticized for selecting a sample from only manufacturing firms located in north India. Doherty, Carcary [26] conducted exploratory research to assess the drivers and barriers of cloud adoption in SMEs in Ireland and found cost-benefit and continuous availability of service providers as a key driver and a key challenge respectively. As this study is exploratory, it is difficult to identify the predictability of factors of cloud adoption. Gupta, Seetharaman [27]

performed another study on small and micro enterprises and discovered that ease of use and convenience, security and privacy, and cost reduction had a significant influence on cloud computing adoption. They did not employ any established IS adoption model and ignored some critical cloud computing-related constructs in their study.

In another study, Alshamaila, Papagiannidis [10] examined the determinants of cloud computing adoption by SMEs in the northeast of England. The use of small sample size (15 SMEs) and qualitative analysis make it difficult to generalize the outcomes of the study. Agostino, Søilen [28] conducted a descriptive analysis to identify the critical success factors for SMEs of cloud-based innovative services. In this study, they have considered the adoption of one cloud-based application among SMEs in the developed country context, which is not much relevant in developing country context. Top management support, firm size, technology readiness, relative advantage, competitiveness, complexity, and compatibility were found as influential factors of cloud computing adoption among SMEs in Kenya [29]. The employment of descriptive design in this study makes it problematic to evaluate the relative importance of factors affecting cloud computing adoption.

Based on the extensive review of literature, this research attempts to fill the research gaps by developing a comprehensive model, at the same time, by using a robust methodology to validate the proposed model. The authors believe that this study will extend the conventional empirical literature, particularly in the domain of information systems research.

2.2. Technology-organization-environment (TOE) framework

At organization-context adoption, Tornatzky, Fleischer [30] proposed the technology-organization-environment (TOE) framework based on Fred Fiedler's contingency theory that groups the common factors that affect technology adoption. The review of prior cloud computing literature (Table 1) suggests that the TOE framework provides an appropriate starting point for studying cloud computing. Low, Chen [31] claim that a theoretical model of cloud computing adoption needs to take into account factors that affect the intention to adopt and utilize cloud computing, which is embedded in the particular technological, organizational, and environmental contexts of an organization. The technology component defines the characteristics of internal and external technologies that might affect individual or organization. The organization context focuses on the obtainability and expertise of the firm in using different resources to assist the acceptance of an information system by an enterprise. This context comprises enterprise size, firm scope, culture, organizational structure, communication channels, level of centralization, management beliefs, and sufficiency of the required human resources. The environment component encompasses different industry factors such as market elements, competitors, and the regulatory environment, which affect directly or indirectly the operations of an enterprise.

TOE framework is consistent with the Rogers [32] diffusion of innovation (DOI) theory as it concentrates on both internal and external conditions of the organization, as well as technological characteristics in study of drivers for new technology diffusion [33]. Since TOE framework considers multidimensional aspects of an organization when studying technology adoption and diffusion, it is perceived that TOE has more explanatory capacity compared to other adoption models such as Technology Acceptance Model (TAM) [34], Theory of Planned Behavior (TPB) [35], the Unified Theory of Acceptance and Use of Technology (UTAUT) [36], Diffusion of Innovation (DOI) [32] in examining technology adoption and its use, particularly at organization context [13]. Existing theoretical and empirical evidence revealed that TOE framework had been a popular foundational model in examining emerging technologies like cloud computing adoption, implementation, and usage [22,24,31]. Awa, Ukoha [15] further elaborate that TOE has been considered as a widely used framework that focuses on people rather on roles and believes that people are not substitutable component. Ramdani, Chevers [37] argue that a large number of the literature generate fragmented findings as researchers investigate the impact of only a limited number of variables, or to pick and choose from a list of factors that have been empirically tested and validated to study adoption/diffusion of ICT. They further stated that the TOE framework is claimed to be an integrative framework that offers a holistic and guiding theoretical basis since the IT/IS adoption/diffusion studies typically evaluate different technological, organizational, and environmental circumstances that facilitate or inhibit adoption/diffusion. Moreover, this framework is organization size and industry restrictions free [23]. Therefore, TOE has been claimed to be a "generic" theory of technology adoption, implementation, and post-adoption diffusion among organizations [24,27,29].

2.3. Extension of TOE framework

Previous scholars have recurrently worked on the factors within TOE's framework to reinforce its theoretical ground and have proved theoretical and empirical conformity [38]. However, Tornatzky, Fleischer [30] hardly claimed TOE as a rigid framework. Venkatesh, Thong [39] claimed that modification, revision, and extension of an individual model are required in certain distinct information technology application, and therefore many IS scholars [38] extended the existing TOE framework by adding different contextual constructs from other dimension or blending it with other theoretical models to improve the predictive capability of this framework. Besides, the addition of a new dimension with existing TOE model depicts a clearer understanding of organizations' acceptance of domain-specific technology [40]. In this spirit of theory advancement, this research proposed an extension to the existing TOE framework.

This paper added individual context in the proposed model as the capabilities of the TOE framework were enriched when individual context, as well as its cognate factors, were integrated [13]. The first time adoption or operational use of any information technology at individual or organization levels is a voluntary decision [41]. In SMEs context, all the strategic decisions are strongly influenced by the owners or managers. Subsequently, a reflection of the personal attributes of the owners or CEOs is seen in the decision making process. Many scholars [13,36,42] affirm that firm's strategies are directly shaped by the perceptual idiosyncrasies of decision-makers as well as their emotional feelings of the technology's usefulness and effortlessness, which ultimately drive motivation and attitude. Often, firm-level technology adoption decision, particularly in SMEs, is influenced by decision--makers' enthusiasm and growth ambition, owners' emotion and motivation managers' belief differences, and CEO's knowledge and characteristics [38,43]. Nevertheless, in the literature, individual characteristics (e.g., attitudes, beliefs, perceptions, psychographics, motivation) were rarely adopted suitably in the TOE framework. Although TOE framework recognizes decision actors' perception as a factor under the context of an organization, considering decision actors as an independent context rather than a construct has the probability of improving the explanatory capacity and predictability of TOE framework [15].

Further, this study added perceived risks, server location, and cloud provider's influence as contextual variables in the proposed research model. The presence of perceived risk towards utilizing technology is significant for enterprises because security and privacy, which increase perceived risk, are considered as major challenges for adopting cloud [24]. The physical location of the server, another important component of cloud computing, is a concern for an enterprise because of the storage of organizations' resources in other countries. Sometimes end users are not informed properly about the exact location of servers which could drive to compliance and privacy concerns [22]. Further, the influence of cloud providers affects organizations to adopt cloud computing [44]. Plentiful supports from the cloud providers may increase cloud computing adopted organizations.

Furthermore, although existing information systems literature

confirmed the positive impact of innovative technology adoption on firm performance, very few studies conducted on cloud computing adoption addressed this post-adoption consequences because researchers excessively emphasized the initial adoption decision [45]. Therefore, this study included firm performance as a second key-dependent construct in the proposed model. Based on above discussion, this study has developed a theoretical framework by integrating individual dimension and different context-specific constructs with the TOE framework. Fig. 1 presents the proposed research model.

2.4. Control variables

In this research, we have included two control variables namely firm size (FS) (e.g., number of employees) and industry (IND) in which SMEs operate to reduce the variance in the firm performance that might be affected by these variables. According to Low, Chen [31], the probability of having higher financial and labor resources by large organizations may lead to variances in firm performance. Likewise, this study includes industry type as a control variable to address the performance variances that might be occurred due to industry-specific physiognomies such as industry environment, market competition intensity [24].

3. Hypotheses development

3.1. Technology context

3.1.1. Relative advantage (RA)

Relative advantage can be defined as the "degree to which an innovation is perceived as being better than the idea it supersedes'' [32]. Enterprises tend to adopt emerging technology that has significant and unequivocal benefits of generating their strategic and operational success [24]. The relative advantage of using cloud computing is that it reduces cost, saves business' time, increases sales, enhances productivity, and offers the opportunity to use shared resources [22,46]. Cloud computing provides SMEs numerous benefits which were not possible in the past, such as pay-per-use option, enhanced scalability and flexibility, and easier installation and upgradation process [9]. Alshamaila, Papagiannidis [10] stated that when SMEs perceive cloud computing providing a relative advantage, then it is more likely that they adopt cloud computing in their operations and processes. They further claimed that cloud scalability and mobility offer them more control over their operations and IT expenditure. The proper deployment of cloud services will help SMEs perform operational tasks more conveniently and efficiently [27]. Thus, it is meaningful to examine whether relative advantage affects the adoption of cloud computing in SMEs.

H1. Relative advantage has a positive impact on cloud computing adoption.

3.1.2. Service quality (SQ)

The level of service quality of the technology to the end-users has significantly affected the use of online services in an organization [22]. In this study, the dimensions of reliability, assurance, and responsiveness have been adopted to evaluate reliability, availability, and speed of delivery of service of cloud computing to the end-users of the organization. In developing countries like Bangladesh where cloud computing is relatively new, the level of service quality is considered as one of the important predictors towards its adoption. Since cloud computing offers services to end-users through the Internet and network technologies at any time and anywhere basis, cloud services must be uninterrupted, fast, functional, and responsive [22]. This study estimates that better service quality will enhance the possibility of adopting cloud services among SMEs. Thus, this research hypothesizes the subsequent:

H2. Service quality has a positive impact on cloud computing adoption.

3.1.3. Perceived risk (PR)

Perceived risk encompasses various forms of financial, social, psychological, physical, or time risks associated with using network-based services [47]. People often show anxiety about conducting transactions online, which negatively influence to accept any online service [48]. Earlier studies identified the direct negative influence of perceived risks on new technology adoption [27,49]. As an emerging technology, cloud computing delivers various services by using the Internet and network technologies. In the cloud environment, one of the major challenges is the concern about the privacy and security of the resources that are shared with the third party. This anxiety further enhances the perceived risk of utilizing cloud services in the organization, which hinders the adoption of cloud technology. There are many plausible risks that SMEs may be worried about, such as deception, lack of data security on the cloud, unauthorized use of cloud data, and other illegal activities [9,24,50]. Therefore, we propose as follows.

H3. Perceived risk has a negative influence on cloud computing adoption.

3.2. Organization context

3.2.1. Top management support (TMS)

As the structures of SMEs are simple and centralized, chief executive officers (CEOs), who are also the owners or the chief managers, typically

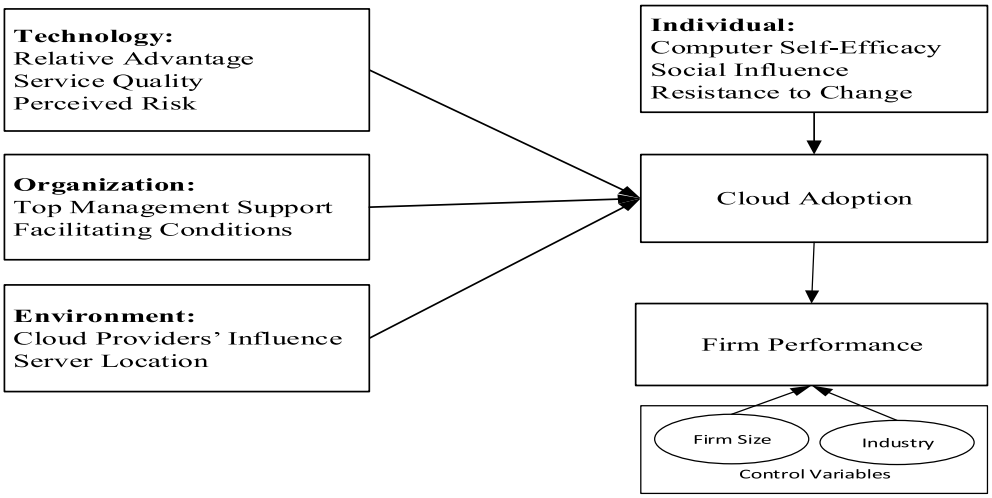


Fig. 1. Proposed research model.

run and manage the business. Hence, top management (CEOs/Owners/Managers) significantly influence IS/IT adoption because they take all decisions including both strategic, tactical and operational. The adoption of cloud computing in SMEs is a strategic decision which is mainly made by top management. In the literature, supports from top management was identified as an important predictor of cloud computing adoption [9,23]. Adequate supports from top management significantly influence the utilization of cloud technology as top management play key roles in allocating different resources, integrating services, and re-engineering business processes [31]. Besides, they encourage the members of the organization to implement the change occurred due to the adoption of new technology. Hence, the acceptance of cloud technology significantly depends on the level of supports that SMEs receive from their top management. The failure of top management in recognizing the potential benefits of cloud technology certainly discourages them from adopting this technology. Hence, this research attempts to measure the effect of top management support on cloud adoption in SMEs by formulating the following hypothesis.

H4. Top management support has a positive influence on cloud computing adoption.

3.2.2. Facilitating conditions (FC)

Facilitating conditions include both the structural aspects and the technical human resources of an organization [51]. The structural aspects refer to the technical infrastructure such as installed network technologies and enterprise systems. On the other hand, the technical human resources refer to the workforces within the organization who have IT knowledge and skill to implement the cloud-computing services. In other words, this factor denotes how well cloud computing matches with the current business processes and corporate philosophy. 24,29. Several empirical studies also revealed that facilitating conditions play as a facilitator of technology adoption [51–53]. In the context of SMEs, facilitating conditions may include IT infrastructure, uninterrupted Internet connection, IT skills, and compatibility with the current business process. In our study, we made a standard classification of facilitating conditions which are listed under organization context. Although some facilitating conditions (e.g., IT infrastructure, internet connectivity) could be grouped under technology context. However, we assume that the existence of facilitating conditions play a significant positive role in adopting cloud technology in SMEs. Hence, we posit the subsequent hypothesis.

H5. Facilitating conditions have a significant impact on cloud computing adoption.

3.3. Environment context

3.3.1. Cloud provider's influence (CPI)

The influence of technology providers plays an essential role in the adoption of emerging and disruptive technology. The adoption of the Internet has already experienced the significant effect of technology providers [44]. For instance, firms like Cisco Systems increased awareness in many organizations about the strategic and competitive need to adapt internet technologies to their business processes and operations. Nowadays, firms are focusing more on technology partners rather than technology suppliers so that they can enjoy the continuous supports and esteemed know-how from their partners [44]. When there is a dependent affiliation between the customer organization and its technology supplier, private support can become even more significant [54]. In the domain of cloud computing, a growing number of high tech companies, led by Amazon, are currently developing various giant server farms and offering different cloud-based business models to their customer firms. Accordingly, key cloud providers (e.g., Apache, EMC or Cisco) adjust their technologies to facilitate customer firms access to the cloud [44]. They are intensively caring firms and providing different cloud computing adoption-related products and services with virtual applications and business software so that customer firms can use resources anytime, anywhere [44]. Many established cloud providers (e.g., Google, IBM, Salesforce) are undertaking different marketing activities to encourage customers adopting cloud services. Various supports from cloud providers are likely to motivate firms in implementing cloud services. Thus, we postulate the following hypothesis.

H6. The influence of cloud providers positively affects cloud computing adoption.

3.3.2. Server location (SL)

Cloud computing has brought a revolution in delivering IT-driven services using the ideas of distributed and parallel computing, which offers a remotely-accessible environment of sharing or accessing various computing resources with high reliability, dynamicity, and scalability [3]. In the cloud environment, resources are stored in the third party's server. The actual location of the server might be in any place in the globe. Cloud providers might have several servers in various locations in different countries. Often, they preserve firms' data and resources in those servers without informing customer firms, which could amplify enterprise's degree of anxiety about security and privacy issues towards utilizing cloud services. Also, the storing of firms' resources in the servers located in another country might conflict with the firms' internal rules and regulation, which restricts firms from adopting cloud services. Additionally, there are still no global laws or monitoring codes for ensuring the safety of cloud data and resources [22]. Thus, the location of the server is an important factor in cloud adoption, which needs to be assessed. Hence, the following hypothesis is posited:

H7. Server location positively affects cloud computing adoption.

3.4. Individual context

3.4.1. Computer self-efficacy (CSE)

According to Bandura [55], "self-efficacy refers to the extent or strength of one's belief in one's own ability to complete a task." Computer self-efficacy assesses the level of confidence that a person has in doing the necessary tasks by utilizing computer-based technology [56]. A person who possesses an advanced level of computer self-efficacy is likely to enjoy more comfortability in acquiring innovative knowledge from technology-based resources [57,58]. In the domain of cloud computing, Yang and Lin [59] stated that computer self-efficacy affects users' perceptions towards the adoption of cloud storage service and its continuous usage in the future. In this study, we focus on the level of self-efficacy of decision-makers in SMEs. As the number of decision-makers in SMEs is limited, often confined with the CEOs, the level of computer self-efficacy of CEOs significantly influences the adoption of a new technology [15]. For example, if the CEOs, who are usually the owners or chief managers, have advanced knowledge related to the utilization of computing resources, which improves their self-efficacy level, they are prone to adopt cloud computing in their businesses. Also, the decision-makers with improved level of computer self-efficacy can understand the potentials of emerging technologies, which leads them to accept those technologies. As cloud computing delivers computer and network-based services, this study assumes that decision maker's higher level of computer self-efficacy will motivate them towards the adoption of cloud computing.

H8. CEOs' Computer self-efficacy has a positive impact on cloud computing adoption.

3.4.2. Social influence (SI)

Venkatesh, Thong [60]) defined social influence as "the degree to which an individual perceives that important others believe that he or she should use the new technology." According to Rashotte [61], "social influence refers to the changes in an individual's thoughts, feelings, attitudes, or behaviors due to communicating with other people or

groups." Previous researches on technology adoption have suggested that social influence directly affects the behavioral intention to adopt online services [51,62]. Although social influence plays a significant role in technology adoption at individual level at voluntary context [52, 63], it might also have significant effect at organization level [36]. In this research context, decision-makers of SMEs can be significantly influenced by other owners, managers, colleagues, customers, trade partners, suppliers, competitors, and other stakeholders. We assume that SMEs intention to accept cloud computing services is greatly affected by the behavioural changes in decision-makers due to the interactions with other individual or groups. Therefore, we hypothesize the subsequent.

H9. Social influence positively affects cloud computing adoption.

3.4.3. Resistance to change (RC)

The adoption of technology involves transforming human behavior and the acceptance of the change. In the Information System (IS) literature, resistance to change indicates the adverse reaction of adopters to the proposed change [64]. Resistance to change is considered to be one of the important factors in the adoption of technology and the failure or success of some new systems and services [65]. For example, Hoque and Sorwar [66] found that resistance to change negatively influence m-Health adoption in the context of Bangladesh. In this research context, as SMEs possess diversity in terms of decision-makers' characteristics (i.e., attitude, personality, age, gender), often they face severe resistance while adopting an emerging technology like cloud computing. The adoption of cloud services may cause transformational changes in the existing organizational structures, systems, operations, and processes. Decision-makers are expected to take the challenges of managing changes effectively and efficiently. Therefore, many decision-makers get fear and feel discomfort while thinking of using new technology like cloud computing. This study believes that resistance to change is one of the inhibitors of the adoption of cloud services in SMEs.

H10. Resistance to change has a negative effect on cloud computing adoption.

3.5. Cloud computing adoption (CA) and firm performance (FP)

Performance is defined as the key output of any strategic endeavor initiated by an organization. The effective adoption and alignment of IT resource to improve the core competencies of a firm enhance firm performance [67]. Earlier researches documented the positive effect of the adoption and utilization of any innovative technology (e.g., EDI, IoT, IOS, RFID) on firm performance. For example, Chan and Chong [16] found that interorganizational systems implementation help organizations to improve operational performances, such as improving flexibility, production lead time, cost-saving, forecasting, resource planning, and inventory level, as well as SCM related performances such as improving costing accuracy and increasing coordination among departments, suppliers and customers, and amplifying revenues. In the domain of cloud technology, performance indicates the potentials of cloud services that can positively affect the cloud supported operations and processes of an organization. The successful adoption of cloud computing significantly affects firm performance. For instance, cloud computing provides the opportunity to access remote IT resources owned by the third party, which reduces the cost of internal IT infrastructure development, IT maintenance, and IT obsolesce cost. Ooi, Lee [5] conducted a study in the context of Malaysian manufacturing companies and found positive impact of cloud computing on non-financial performances (e.g. flexibility and quality). Cloud computing, as newly available IT resource, enables SMEs to concentrate on core business capabilities, enhanced operational performance, productivity and financial performance. For instance, Garrison, Wakefield [45] stated that the successful implementation of cloud computing positively influences the organizational performances for cloud-supported operations. Therefore, we formulate the following hypothesis.

H11. The adoption of cloud computing positively influences firm performance.

4. Methodology

4.1. Variables and measurements

A survey questionnaire, divided into two segments, was designed based on the recommendations generated from conducting a focus group discussion with five professionals. The first segment contained 43 close-ended questions to measure the 12 constructs used in the model. The items representing each construct were adapted from literature, and some of them were revised according to the research context (seeAppendix A). We used five-point Likert scale ranging from 1 to 5, where 1 = strongly disagree and 5 = strongly agree, to design the closed-ended questions. The second segment dealt with demographic information. We measured the control variables using the method used in the literature [45]. Authors conducted a pilot test using 30 samples, who were asked to participate voluntarily. We made some minor revisions of the research questions based on the results of the pilot test. Some owners/CEOs claimed to use more familiar and context-specific words in designing questions as they found them more convenient to respond. In response to their concerns, we changed relatively infrequent words with the most common words on the condition that such change would not influence construct's content validity. We have collected ethical approval from the Center for Modern Information Management, Huazhong University of Science and Technology, Wuhan, China and Institute of Health Economics, University of Dhaka, Dhaka, Bangladesh.

4.2. Sampling and data collection

The target population of this study was SMEs which dominate the private sector and play an indispensable role in the economic development of Bangladesh. This research chose Bangladesh that has a rapidly growing economy so that it can be considered as an acceptable example of a developing country particularly in the East Asia region [68]. The survey was administered among SME's owners/CEOs as they play a key role in the decision-making process. Respondents were selected based on their relevant knowledge and willingness to respond. We applied a non-probability sampling technique, namely the accidental sampling in which respondents answered the questionnaire according to their availability and willingness [69]. We selected this technique as it was quicker and easier to manage compared to the other techniques that needed more resources (e.g. technical resources, time). Also, accidental sampling is a common method used in the data collection process in technology adoption research [70]. The survey was conducted in the months from June 2018 to September 2018. We distributed a total of 450 questionnaires to the respondents, among them 335 questionnaires were returned resulting 74.44% response rate. After that we prepared a dataset consisting of 335 records using SPSS software package. As the dataset contained several corrupt or inaccurate records (e.g., duplicate observations, irrelevant observation, structural errors, incomplete records, and missing data), we performed data cleansing method to correct the data set. We fixed some structural errors (e.g., typos or inconsistent capitalization and mislabelled classes) as well as removed duplicate, irrelevant and missing records. After conducting data cleansing process, a total of 311 questionnaires were assembled. The demographic profile of our samples is depicted in Appendix B. Most of the firms had been carrying out their business operations for more than ten years. The largest 35.7% of organizations had employees with 100–200. Moreover, service sector (52.4%) dominated the representation of participating firms while manufacturing sector represented 47.6%.

4.3. Assessment of multivariate assumptions

When the data set is ready for the analysis, we need to perform

several statistical tests to assess the dataset in order to confirm that certain multivariate assumptions are satisfied [5]. First, we tested the normality using Pearson's Skewness and Kurtosis parameters. The acceptable range of these parameters is between −2.58 to +2.58. The results (Appendix C) indicate that all values of the variables were within the acceptable range.

Next, we checked the linearity of relationships among the constructs by utilizing the p-value of the deviation from linearity with ANOVA. The results of the test (see Table 2) specified that there was a combination of linear and non-linear relationships between the predictors and target construct. Hence, we further examined the non-linear relationships by applying Ordinary Least Squares (OLS) test. The p-values (see Table 3) of OLS test indicated that these relationships are linear.

Then, we assessed the multicollinearity problem by using correlation coefficients and variance inflation factor (VIF) values. The results, as shown in Table 5, indicated that all the values for correlation coefficients are lower than the threshold of 0.90 [71]. Besides, the VIF values (see Appendix D) for the constructs were less than the threshold of 10 [72]. Thus, we concluded that there was no multicollinearity problem [5].

Finally, scatter plot was used to examine the homoscedasticity of distribution of the data [73]. Based on the distribution of the regression standardized residuals, this study observed that all residuals are evenly scattered along a straight line, which indicates that the homoscedasticity of the distribution is supported. Hence, we can conclude that all multivariate assumptions have been satisfied.

4.4. Non-response bias

Similar to Ooi, Lee [5], we examined the differences between all key variables used in the research model using the independent *t*-test and the result indicated that there are no significant differences identified. For further verification, we conducted chi-squared test for independence, and the result showed that there are no significant differences. Thus, non-response bias was not found in the dataset used in this study.

Table 2
Deviation from linearity test.

ANOVA Table	Sum of squares		df	Mean Square	F	Sig.	Linear
CA * RA	Deviation from linearity	6.158	14	.440	1.285	.215	Yes
CA * SQ	Deviation from linearity	12.348	15	.823	3.149	.000	No
CA * PR	Deviation from linearity	13.393	12	1.116	2.490	.004	No
CA * TMS	Deviation from linearity	9.878	11	.898	2.105	.020	No
CA * FC	Deviation from linearity	17.741	14	1.267	5.180	.000	No
CA * CPI	Deviation from linearity	4.572	10	.457	1.050	.401	Yes
CA * SL	Deviation from linearity	14.384	14	1.027	4.085	.000	No
CA * CSE	Deviation from linearity	13.550	10	1.355	4.004	.000	No
CA * SI	Deviation from linearity	17.275	13	1.329	4.177	.000	No
CA * RC	Deviation from linearity	22.486	10	2.249	4.731	.000	No
FP * CA	Deviation from linearity	7.532	11	.685	3.425	.000	No

Table 3
Ordinary Least Square (OLS) test.

Model		Sum of Squares	df	Mean Square	F	Sig.
1	Regression	114.350	1	114.350	396.103	.000[b]
	Residual	89.204	309	.289		
a. Dependent Variable: CA						
b. Predictors: (Constant), SQ						
1	Regression	57.054	1	57.054	120.339	.000[b]
	Residual	146.500	309	.474		
a. Dependent Variable: CA						
b. Predictors: (Constant), PR						
1	Regression	66.558	1	66.558	150.125	.000[b]
	Residual	136.996	309	.443		
a. Dependent Variable: CA						
b. Predictors: (Constant), TMS						
1	Regression	113.650	1	113.650	390.612	.000[b]
	Residual	89.904	309	.291		
a. Dependent Variable: CA						
b. Predictors: (Constant), FC						
1	Regression	114.965	1	114.965	401.002	.000[b]
	Residual	88.589	309	.287		
a. Dependent Variable: CA						
b. Predictors: (Constant), SL						
1	Regression	88.823	1	88.823	239.225	.000[b]
	Residual	114.731	309	.371		
a. Dependent Variable: CA						
b. Predictors: (Constant), CSE						
1	Regression	92.106	1	92.106	255.370	.000[b]
	Residual	111.449	309	.361		
a. Dependent Variable: CA						
b. Predictors: (Constant), SI						
1	Regression	38.972	1	38.972	73.169	.000[b]
	Residual	164.582	309	.533		
a. Dependent Variable: CA						
b. Predictors: (Constant), RC						
1	Regression	42.349	1	42.349	194.999	.000[b]
	Residual	67.107	309	.217		
a. Dependent Variable: FP						
b. Predictors: (Constant), CA						

Table 4
Reliability and convergent validity.

Constructs	Item loading range	Cronbach's Alpha	rho_A	Composite Reliability	AVE
CA	0.866–0.903	0.865	0.865	0.918	0.788
FC	0.825–0.863	0.860	0.866	0.905	0.704
FP	0.740–0.823	0.774	0.774	0.856	0.597
PR	0.792–0.837	0.826	0.832	0.884	0.656
RA	0.810–0.835	0.843	0.845	0.894	0.679
RC	0.813–0.830	0.764	0.766	0.864	0.679
CSE	0.875–0.900	0.865	0.866	0.918	0.788
SI	0.839–0.852	0.864	0.864	0.907	0.710
SL	0.808–0.851	0.848	0.849	0.898	0.687
SQ	0.801–0.887	0.858	0.863	0.904	0.702
TMS	0.831–0.895	0.816	0.822	0.890	0.730
CPI	0.753–0.869	0.759	0.778	0.861	0.674

Note: AVE = Average Variance Extracted.

4.5. Common method bias

Common method bias (CMB) is intrinsic to behavioral study which uses cross-sectional survey data [74]. As all the data is self-reported, we examined CMB that could be a future risk to the results of the study. To assess CMB, we used Harman's single-factor test by using SPSS 23.0 version software. To identify the number of factors critical for the description of variance, we applied principal axis factor analysis (PAF) [75]. The results of our test indicated that a single construct was

Table 5
Fornell-Larcker criterion.

	CA	FC	FP	PR	RA	RC	CSE	SI	SL	SQ	TMS	CPI
CA	0.888											
FC	0.751	0.839										
FP	0.623	0.651	0.773									
PR	0.535	0.710	0.523	0.810								
RA	0.692	0.677	0.549	0.649	0.824							
RC	0.441	0.528	0.481	0.545	0.445	0.824						
CSE	0.661	0.577	0.492	0.476	0.601	0.350	0.888					
SI	0.672	0.634	0.554	0.627	0.586	0.500	0.552	0.842				
SL	0.752	0.627	0.587	0.519	0.579	0.503	0.568	0.700	0.829			
SQ	0.752	0.723	0.594	0.617	0.681	0.497	0.650	0.719	0.693	0.838		
TMS	0.584	0.520	0.458	0.379	0.416	0.375	0.370	0.411	0.479	0.483	0.854	
CPI	0.586	0.598	0.538	0.579	0.538	0.568	0.458	0.526	0.539	0.578	0.407	0.821

responsible for 38.42% of the total variance which is far below than the recommended 50% [74]. Also, we examined the correlation matrix of the constructs (Table 5) in the measurement instrument and found that there was no correlation greater than 0.90. Further, we tested CMB based on variance inflation factors (VIF) values (see Appendix D) of the constructs generated from full collinearity test. These values were below than the suggested 3.3 [76]. Thus, CMB was not a potential threat in this study.

4.6. Data analysis

At the beginning stage, using SEM, we confirmed the reliability and validity of the constructs and evaluated the predictive relevance of the model, and the total variance explained. Then, we ran importance-performance map analysis (IPMA) in PLS to examine the importance of the independent variables and their performance in explaining the dependent variable. In this study, variance-based Partial Least Squares (PLS) Structural Equation Modelling (SEM) approach was considered as the most appropriate approach as it has the ability to examine a series of relationships simultaneously, whereas other statistical methods (e.g., multiple regression, or multivariate analysis of variance) are confined to analyzing the relationships between each construct singly [77]. Besides, the nature of this study better suits PLS-SEM as this research focuses on theory development or theory exploration rather than theory confirmation. The complexity of the model and a large number of indicators also lead to the use of variance-based PLS-SEM. Finally, as this study aims to enhance the variance explained, which is data-driven rather than estimation of model fit, we used variance-based PLS-SEM in place of covariance-based SEM (CBSEM). However, SEM is unable to examine the non-linear relationships among the constructs [78], which might cause the actual complex decision-making process to be analyzed simply [20]. In this study, there were non-linear relationships that exist among the constructs in the model (see Tables 2 and 3). Therefore, the ANN approach is requisite to supplement the SEM analysis [5]. At the final stage, we performed neural-network analysis to corroborate the SEM results as well as to prioritize the key predictors of cloud adoption. In sum, at the first stage, SEM was conducted to determine the statistically significant predictors, while at the second stage, ANN analysis was performed to prioritize the predictors based on their relative importance in explaining their target constructs.

5. Results

5.1. Measurement model

We used internal reliability, convergent validity, and discriminant validity criteria to assess the measurement model [79]. The common measures to evaluate the reliability of the constructs are Cronbach's Alpha and Dijkstra-Henseler's rho (ρA). In this study, the value of reliability criteria was greater than the suggested value of 0.70 [80] (see Table 4).

We measured convergent validity by checking the values of item loadings, composite reliability, and average variance extracted (AVE). It was evident from the results, as depicted in Table 4 that the values of item loadings and AVE are above the recommended values of 0.50 [81]. The values of composite reliability were also above the suggested value of 0.70 [80]. Likewise, we examined discriminant validity by checking the correlations among the measures of possibly overlying variables. In this paper, the value of the square root of AVE for each variable was above the correlation of that variable with other variables (see Table 5). Additionally, all values of HTMT ratio, as shown in Table 6, were <0.85. These measures confirmed the discriminant validity of the studied constructs [82].

5.2. Structural model

The goodness of the structural model is assessed by the R^2 that measure the coefficient of determination and the significance level of the path coefficients [79]. From the results of our study, the adjusted R^2 value for cloud adoption was 0.772, that means, 77.2% of the variance

Table 6
Heterotrait-Monotrait ratio (HTMT).

	CA	FC	FP	PR	RA	RC	CSE	SI	SL	SQ	TMS	CPI
CA												
FC	0.846											
FP	0.762	0.793										
PR	0.627	0.841	0.651									
RA	0.807	0.794	0.679	0.773								
RC	0.540	0.649	0.626	0.686	0.553							
CSE	0.764	0.668	0.601	0.562	0.701	0.428						
SI	0.777	0.732	0.676	0.741	0.684	0.612	0.637					
SL	0.835	0.730	0.723	0.616	0.682	0.623	0.663	0.816				
SQ	0.842	0.842	0.728	0.730	0.800	0.615	0.755	0.834	0.810			
TMS	0.686	0.606	0.565	0.446	0.485	0.455	0.435	0.478	0.567	0.564		
CPI	0.716	0.736	0.697	0.730	0.661	0.754	0.563	0.645	0.662	0.707	0.500	

Table 7
Hypotheses testing.

Hypotheses	Path	β	t- value	P-value
H1	**RA - > CA**	0.169	3.843**	0.000
H2	**SQ - > CA**	0.119	2.802**	0.005
H3	**PR - > CA**	−0.169	3.752**	0.000
H4	**TMS - > CA**	0.156	3.879**	0.000
H5	**FC - > CA**	0.283	5.106**	0.000
H6	**CPI - > CA**	0.089	2.277*	0.023
H7	**SL - > CA**	0.265	5.079**	0.000
H8	**CSE - > CA**	0.131	3.700**	0.000
H9	**SI - > CA**	0.087	1.795	0.073
H10	**RC - > CA**	−0.083	2.422*	0.015
H11	**CA - > FP**	0.626	14.881**	0.000
Control Variables	**IND - > FP**	−0.078	1.681	0.086
	FS - > FP	0.022	0.476	0.634

Adjusted R^2 for CA = 0.772, Adjusted R^2 for FP = 0.397 *Significant at $p < 0.05$ **Significant at $p < 0.01$.

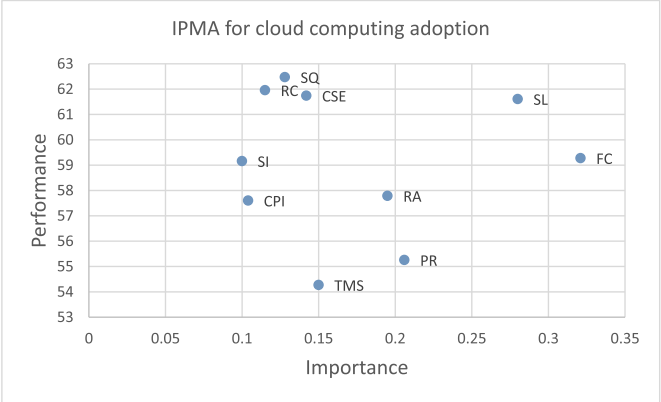


Fig. 2. IPMA for cloud computing adoption in SMEs.

in cloud adoption can be explained by the factors. Similarly, the adjusted R^2 value for firm performance was 0.397, that implies, 39.7% of the variance in performance can be explained by the cloud computing adoption. Therefore, the proposed model is statistically confirmed.

This study investigated the relationships between endogenous variables and exogenous variables using path coefficient (β) and t-statistics. The results of all hypotheses testing are summarized in Table 7. The results indicated that hypotheses H1 (RA - > CA, β = 0.169, t = 3.843), H2 (SQ - > CA, β = 0.119, t = 2.802), H3 (PR - > CA, β = −0.169, t = 3.752), H4 (TMS - > CA, β = 0.156, t = 3.879), H5 (FC - > CA, β = 0.283, t = 5.106), H6 (CPI - > CA, β = 0.089, t = 2.277), H7 (SL - > CA, β = 0.265, t = 5.079), H8 (CSE - > CA, β = 0.131, t = 3.700), and H10 (RC - > CA, β = −0.083, t = 2.422) were statistically significant. The results imply that the hypothesized relationships between relative advantage, service quality, perceived risks, top management support, facilitating conditions, cloud provider's influence, server location, computer self-efficacy, resistance to change and cloud computing adoption in SMEs are empirically confirmed. In contrast, the p-values (>0.05) for the path for hypothesis H9 indicated that social influence had an indirect effect on the adoption of cloud computing. The path for hypothesis H11 (CA - > FP, β = 0.626, t = 14.881) was also statistically supported, which confirmed the strong and positive relationship between cloud computing adoption and firm performance. The influence of two control variables (e.g., organization size, industry) was found statistically insignificant, which indicated that these two variables did not cause for significant variance in firm performance.

5.3. Importance-performance map analysis (IPMA)

We conducted importance-performance map analysis (IPMA) in PLS using cloud computing adoption as the outcome construct. According to Ringle and Sarstedt [83], "The IPMA gives researchers the opportunity to enrich their PLS-SEM analysis and, thereby, gain additional results and findings." The objective of running IPMA is to estimate the importance of the predecessor constructs and their performance in describing the target construct (cloud adoption). This analysis recognizes factors that have a relatively high path coefficient (importance) for the target variable but a relatively low performance [77]. In IPMA, marginal effects of predecessor constructs represent the importance of shaping target construct, while the average latent variable scores of predecessor constructs represent their performance [84]. The performance scores were computed by rescaling the latent constructs scores to a range of 1–100. Table 8 shows the values of the predecessor constructs' importance and performance scores.

Table 8
Importance-performance map analysis.

Latent constructs	Path coefficient (Unadjusted)	Construct performance
FC	0.321	59.274
PR	−0.206	55.253
RA	0.195	57.784
RC	−0.115	61.954
CSE	0.142	61.738
SI	0.100	59.156
SL	0.280	61.607
SQ	0.128	62.470
TMS	0.150	54.267

Further, we developed a priority map (as shown in Fig. 2) by plotting the predecessor constructs' importance and performance scores. From the map, it is observed that server location, facilitating conditions, perceived risk, relative advantage, and top management support are very important factors in determining cloud computing adoption because they have relatively higher path coefficient (importance) compared to other factors in the proposed model. Nonetheless, the performance of facilitating conditions, perceived risk, relative advantage, and top management support are lower than other factors (service quality, resistance to change, and computer self-efficacy). On the other hand, the performance scores of service quality, resistance to change, and computer self-efficacy are higher than other factors despite their lower importance scores. From IPMA, we can conclude that relative advantage, perceived risk, and top management support need special managerial consideration to enhance the adoption of cloud computing in SMEs. On the other hand, managers need to reduce their excessive focus on service quality, resistance to change, and computer self-efficacy as they are less important in describing cloud computing adoption. In sum, managers should allocate more resources to reduce perceived risk, provide relative advantage, and ensure top management support.

5.4. Neural-network analysis

Artificial neural network (ANN) technique resembles the structure of the human brain comprising the neuron, synapse, and the axon. It can enrich knowledge through learning process [5]. According to Haykin [85], "ANN is a massively parallel distributed processor made up of simple processing units, which have a natural propensity for storing experimental knowledge and making it available for use." Neural networks possess the competency to address nonlinearity and to learn by building input-output mapping. The key benefit of employing neural networks is to measure the complicated linear and non-linear relationships between predictors and the acceptance decision and to prioritize the factors based on their relative importance [16]. Moreover, neural networks are more robust and have greater prediction accuracy than the usual regression models [16,50,70].

5.4.1. Validations of neural network

This research performed neural network analysis by using SPSS 23.0

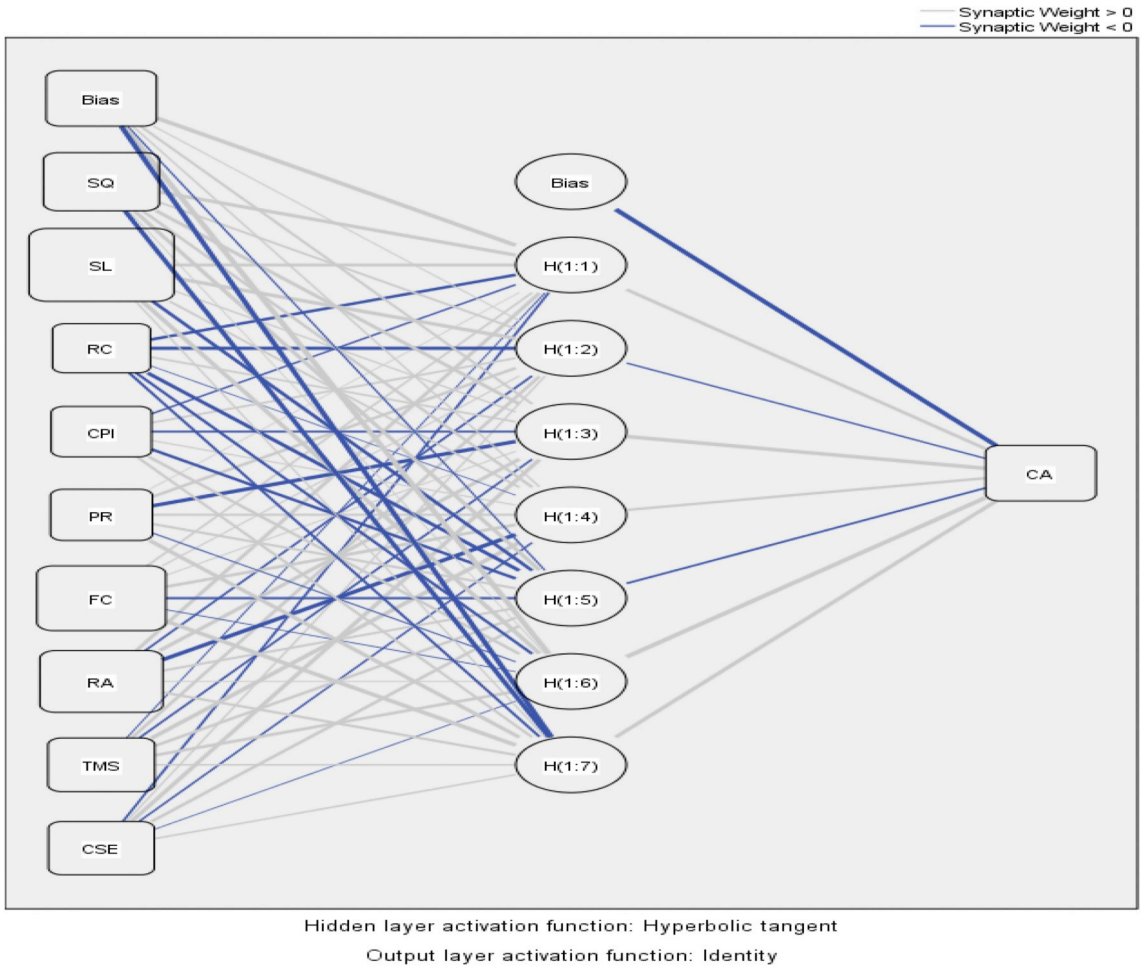


Fig. 3. Neural network model.

Table 9
Root Mean Square Error (RMSE) for neural network model.

Network	Training	Testing
ANN1	0.145	0.055
ANN2	0.069	0.037
ANN3	0.186	0.052
ANN4	0.190	0.032
ANN5	0.154	0.086
ANN6	0.166	0.105
ANN7	0.206	0.059
ANN8	0.086	0.022
ANN9	0.214	0.110
ANN10	0.189	0.156
Mean	0.160	0.071
Standard deviation	0.046	0.040

Table 10
Independent variable importance.

Factors	Relative importance	Normalized Importance
SQ	0.117	57.6%
SL	0.202	100.0%
RC	0.064	31.7%
CPI	0.071	35.3%
PR	0.076	37.7%
FC	0.155	76.5%
RA	0.138	68.0%
TMS	0.090	44.4%
CSE	0.087	43.3%

software package. The neural-network model was developed employing multilayer perceptron training method. This research employed one hidden layer since a continuous function can be represented adequately by one hidden layer [53]. We applied the ten-fold cross-validation technique to overcome the overfitting problem. In this study, we allocated 70% data for training the neural network model and the rest 30% data for testing the prediction precision of the trained model. The nine significant constructs namely relative advantage, service quality, perceived risks, top management support, facilitating conditions, cloud provider's influence, server location, computer self-efficacy, and resistance to change were used in the input layer and cloud computing adoption was used in the output layer (see Fig. 3). The average cross-validated Root Mean Square Error (RMSE) values for training and testing were 0.160 and 0.071 respectively (see Table 9). These low RMSE values denote that the neural network model is fairly consistent in explaining the numerical relationships between the predictors and output.

5.4.2. *Ranking of factors*

We ran a sensitivity analysis in the neural network model to prioritize the predictors by detecting their relative importance. According to Chong [19], "the relative importance of the predictors is a measure of the degree to which the network's model predicted changes in the output values concerning the changes in values of independent variables." According to normalized variable importance values, as shown in Table 10, the most important predictor of cloud computing adoption is server location followed by facilitating conditions, relative advantage, service quality, top management support, computer self-efficacy,

perceived risks, cloud provider's influence, and resistance to change.

6. Discussion

6.1. Technology context

The relative advantage was found as an important facilitator of cloud adoption in SMEs. Our findings contradict with the findings of Alkhater, Walters [22] and Low, Chen [31], who stated that organizations are not still aware of the relative advantage of utilizing cloud computing. In contrast, the findings of Oliveira, Thomas [24], Alshamaila, Papagiannidis [10], and Makena [29] supported our findings. They claimed that firms recognize the relative advantages of adopting cloud computing. Our study reinforces that SMEs primarily focus on the potential benefits (e.g., increase operational efficiency, increase productivity, reduce IT cost, and generate new business opportunities) that can be gained by utilizing cloud technology, instead of using traditional technology. Also, this study observed the positive effect of service quality on cloud adoption. As cloud computing is a network based technology offered by a third party, SMEs expect good service quality to the end users. The study of Alkhater, Walters [22], conducted in Saudi Arabia context, has supported this finding. They further claimed that the relationship between service quality and cloud computing adoption is stronger in SMEs compared to large organizations. SMEs lack necessary technical resources, IT skills, and experience to handle any failure in cloud services, therefore they are driven by the service quality to use cloud services. SMEs prefer to use a simple technology, which requires less efforts, minimum technical knowledge, and less time. Thus, the outcomes of this research imply that confirming good service quality in the service level agreement (SLA) is an essential step to escalate the adoption of cloud computing.

Conversely, perceived risk was found as an inhibitor of cloud computing adoption in SMEs in the Bangladeshi context. This result is consistent with some former researches conducted on the acceptance of technology-based services. For instance, Lancelot Miltgen, Popovič [86] revealed that there is a negative relationship between perceived risks and behavioral intention to adopt biometrics technology. Similarly, Johnson, Kiser [87] found that perceived risks play a negative role towards the acceptance of mobile payment services. However, our study contrasts with the research of Veeramootoo, Nunkoo [48] who concluded that perceived risks have a negative but insignificant effect on the success of an e-government service in the context of Mauritius. However, our findings describe SME's concerns about the breach of security and privacy related to the use of cloud services. Lack of trust towards cloud providers may also increase the perception of risk. Therefore, ensuring security and privacy and building trust among the involved parties are the major challenges for cloud implementation, particularly in developing countries.

6.2. Organization context

This study offers pragmatic evidence that various supports from top management (allocating resources and engaging themselves in the process) are vital for accelerating the adoption of any promising technology like cloud computing at the organization level. This finding is consistent with the outcomes of Ahani, Ab Rahim [17] and Kumar, Samalia [9], who confirmed that top management support is the most significant organizational factor that affects the adoption of new technology. In today's competitive business environment, SMEs need top management to support them in every possible way [17]. Without substantial supports from top management (e.g. owners/CEOs, managers/IT managers), SMEs are unlikely to adopt cloud computing [10]. It is expected that SMEs receive necessary supports from top management more easily as they often have a simple management structure compared with large organizations [88]. Similarly, this study revealed that facilitating conditions had a strong and positive effect on the adoption of cloud computing. Both IPMA and neural network results identified that facilitating conditions as a critical predictor of cloud adoption in SMEs in Bangladesh. This finding is in line with several prior research in the domain of IS literature [51–53]. This study advises that the presence of facilitating conditions such as compatibility with existing processes and systems, technological infrastructure, improved network technologies, high-speed Internet, enterprise application systems, and technical human resources motivates SMEs towards utilizing cloud technologies.

6.3. Environment context

The influence of cloud providers plays a direct and positive role in adopting cloud services, particularly in the context of SMEs. This result implies that cloud providers can support SMEs as they suffer from insufficient financial, technological, and skilled human resources for developing and maintaining their infrastructure. These supports from cloud providers eventually encourage SMEs to adopt cloud services [44]. SMEs' cloud adoption is greatly influenced by the way cloud providers develop trust by providing good services and maintaining collaboration with customers firms because nowadays, SMEs expect technology partners rather technology suppliers. Notably, the concern about the location of servers was found as the most significant determinant of cloud adoption by SMEs in Bangladesh. This finding suggests that SMEs are very concerned about the exact location of their resources. Essentially, such apprehensions may occur since cloud providers store SME's resources in different data repository located in different countries; sometimes, they do not even inform their customers the exact location of servers properly, which could be inconsistent with the privacy and compliance issues for an organization [22]. Therefore, this study suggests that cloud providers should be careful while locating the servers in order to address the issues mentioned above.

6.4. Individual context

In this study, computer self-efficacy of decision-makers had been emerged as an influential determinant of cloud computing adoption in SMEs. This outcome is supported by several prior studies [59,70], which found that computer self-efficacy significantly influence the adoption of innovative technology in different contexts. This finding implies that a high confidence level in effectively using a multifaceted system positively influences the self-motivation and perseverance to adopt cloud computing. Besides, this research found a positive but insignificant effect of social influence on the adoption of cloud computing, which suggests that owners/managers of SMEs of Bangladesh give more focus on benefits rather social influence. Conversely, this study claimed that resistance to change is a significant inhibitor of cloud computing adoption in SMEs. This outcome infers that the decision-makers are conformist to embrace the changes within the organization. In Bangladesh, owners/managers of SMEs usually prefer offline services over IT-based services [68]. Moreover, decision-actors, many of them are elderly, are uncomfortable with the usage of complex technologies. They have lower adaptability with the changes due to the acceptance of new technology. Digital divide and limited technical skills of the decision-makers also increase the propensity to resist the changes associated with cloud adoption.

Following the results of importance-performance map analysis, decision-makers need to focus more on improving the performance of relative advantage, perceived risk, and top management support to increase the adoption of cloud computing in SMEs. Thus, managers need to assign additional resources in delivering more benefits, ensuring data security and privacy, and receiving sufficient supports from top management. In contrast, they need to reduce excessive attention given to service quality, resistance to change, and computer self-efficacy.

Table 11
Comparison between PLS-SEM and neural network results.

Predictors	PLS-SEM		Neural network	
	Path coefficient (Unadjusted)	Ranking	Relative importance	Ranking
RA	0.195	4	0.138	3
SQ	0.128	7	0.117	4
PR	−0.206	3	0.076	7
TMS	0.150	5	0.090	5
FC	0.321	1	0.155	2
CPI	0.104	9	0.071	8
SL	0.280	2	0.202	1
CSE	0.142	6	0.087	6
RC	−0.115	8	0.064	9

6.5. Cloud computing adoption and performance

In line with several previous studies [5,16,45], which observed that the adoption of any IT helps to gain firm performance, this study found the significant impact of cloud computing adoption on SMEs' performance for cloud supported processes and operations. This finding is not astonishing because organizations migrate to the cloud to save cost and achieve performance [45]. The results imply that the successful adoption of cloud computing will help SMEs to gain various strategic and operational benefits, such as increasing annual revenue, minimizing operational cost, and improving flexibility [89]. The proper utilization of cloud services would offer SMEs with the access to state of the art IT resources, therefore enabling enterprises to utilize crucial resources effectively and concentrate on core business operations. SMEs can also improve their IT capability by leveraging the IT competence of the vendor and addressing best practices.

6.6. Comparison between SEM results and neural network results

In this section, we have made a comparison (as shown in Table 11) between results obtained from PLS-SEM analysis and neural network analysis to identify the similarities and differences in determining the importance of predictors. In SEM, predictors' importance was determined by the total unstandardized effects of each independent construct on shaping the dependent construct. According to SEM results, the most significant determinant of cloud computing adoption was facilitating conditions, while the least significant factor was the cloud provider's influence. In contrast, based on independent variable importance scores, neural network analysis identified server location as the most important, and resistance to change as the least important factor in predicting cloud computing adoption. However, top management support and computer self-efficacy were ranked equally in both SEM and neural network results. These variations in results of two different methods indicate the superiority of using machine learning tool such as neural network that can measure non-linear relationship and has higher predictive capability over conventional statistical method such as PLS-SEM that can only analyze the linear relationship [16]. This finding provides additional insights for the practitioners about the relative importance of the predictors in explaining cloud computing adoption.

7. Implications

7.1. Theoretical implications

This study extends the empirical literature in the domain of IS, cloud computing in particular. This study has offered a comprehensive model that incorporates technology, organization, environment, and individual dimensions while assessing cloud computing adoption in the context of SMEs. The findings of this study justified the impact of decision-makers' individual attributes (computer self-efficacy, resistance to change) in technology adoption decision in SMEs. Also, this study has explored some new and relevant cloud computing-related constructs such as server location, cloud provider's influence, and perceived risks, which are not widely assessed in the previous researches. This study has further revitalized the significant positive relationship between cloud computing adoption and SMEs' performance that are affected by cloud supported operations.

The inclusion of the individual dimension and several contextual constructs with the original TOE framework enhanced the explanatory power of the proposed model. The proposed model explained 77.2% of the variance of cloud computing adoption and 39.7% of the firm performance in SMEs context, which indicated that the model had a strong and significant predictive capacity. According to Hair, Ringle [90], the R^2 values of 0.75, 0.50, or 0.25 for endogenous variables indicate significant, moderate, or weak coefficients of determination. This study offers not only the extension of the original TOE framework in the context of cloud computing but also the generalizability; thus, this model can be used to assess the adoption intention of any other technology. We have tested the validity and reliability of the scales used in the survey instrument.

Additionally, this research has performed importance-performance map analysis (IPMA) in PLS to examine the factors' importance in determining cloud adoption and also to identify the performance of those factors. IPMA identified several critical factors (e.g., perceived risk, relative advantage, and top management support) that have higher importance but lower performance. These factors need special managerial concentration in terms of allocating resources to accelerate cloud computing adoption in SMEs. Very few studies performed such analysis, particularly in the context of cloud adoption in SMEs.

Finally, this research has employed a hybrid approach by integrating SEM and neural network to validate the proposed model and to prioritize the factors influencing cloud adoption by identifying the relative importance of each significant construct. This study has further made a comparison between PLS-SEM ranking and neural network ranking of the predictors of cloud computing adoption and identified some differences. These differences may occur due to the nature of two different methods; SEM measures linear relationship and ANN measures nonlinear relationship among predictors and target variables. Previous scholars claimed that neural network analysis has more prediction accuracy than SEM [16,21,70]. Nevertheless, this study did not ignore the validity and necessity of traditional statistical methods, which offered a solid foundation in earlier IS adoption studies. This research suggests reinterpreting the previous works on IS adoption by combining linear and non-linear methods, which provides great strength to the existing technology adoption literature.

7.2. Practical implications

7.2.1. Implications for cloud providers and practitioners

SMEs are the emerging market for the cloud providers as it is considered as the life-blood of the economy, especially for developing countries. Most of the business firms of a developing country belong to SMEs. Therefore, this study has significant implications for cloud providers and technology practitioners and will assist them in recognizing the influential factors of cloud adoption. The outcomes of this research signify that the relative advantage, location of the servers, service quality, perceived risks, and the influence of cloud providers are essential for the adoption of cloud computing in SMEs. Cloud computing is an emerging technology, which is still considered as disruptive. Firms are not well aware of the potentials of utilizing cloud services particularly in developing countries like Bangladesh. Therefore, cloud providers should undertake various measures to increase the awareness of the benefits of this technology through various promotional workshops and seminars. They should focus on functional utilities as well as easy-to-use interfaces while designing cloud services for SMEs so that users with limited technological knowledge can easily use these services. Also, cloud providers have to offer a clear instruction or navigation system to

guide users in SMEs about the smooth operation of the services, which would increase the confidence in utilizing cloud technology.

This study has identified that firms in Bangladesh suffer from various supports from cloud providers end, which is hindering the adoption of cloud. Consequently, there is a need for more effective efforts from cloud providers in order to expand cloud adoption in Bangladesh. Further, cloud providers must undertake adequate security and privacy actions to save firms from any undesirable cases of cyber-fraud and security breaches. Importantly, this research suggests that the location of the servers is the most important concern for the SMEs in adopting cloud services. Cloud providers must inform the organizations about the exact location of the servers. The cloud providers should consider the location of the data servers to be inside Bangladesh, which will help SMEs to comply with their regulations and, also play a significant part in lessening perceived risk towards adopting cloud services. Overall, the outcomes of this research will improve cloud providers' existing knowledge about why cloud adoption is still low in Bangladesh; thus, they can make necessary efforts to foster the adoption of cloud services.

7.2.2. Implications for owners/managers

This study reinforces that supports from both top management and IT department are crucial for the adoption of cloud computing at the enterprise level. Also, facilitating conditions such as technology infrastructure, IT knowledge, compatibility with the existing process has been found as the influential antecedent of cloud adoption in Bangladesh. Therefore, CEOs/managers should focus on developing these organizational resources to gain the maximum benefit of cloud technologies. Also, top managers must have the capacity to assess cloud technology and to assimilate that technology into the existing IT infrastructure effectively and efficiently. One key finding of this research is that the individual attributes of owners/managers play a substantial role in the decision of cloud adoption by SMEs in Bangladesh. Decision-makers' computer self-efficacy encourages cloud adoption, whereas, resistance to change inhibits its adoption. Therefore, they should improve their confidence level towards utilizing cloud services in the organization. Similarly, the attitude related to the reluctance to accept the change due to the adoption of cloud services should be improved to grasp the genuine benefits provided by emerging technology. In sum, the outcomes of this research will assist managers in assessing the features of cloud technology, organization, environments, and their personal attributes while deciding if they adopt cloud technology or not. Furthermore, owners/managers can apply this proposed framework to investigate other IT/IS adoption process.

8. Limitations and future research direction

This study has several limitations which lead to the future agenda of research. First, we collected data from a single country, which indicated that this research reveals only the state of Bangladesh. Hence, this research suggests that future research can be conducted to validate the model using a sample taken from other similar countries. Second, we developed the research model by taking several important constructs from four dimensions; further studies can extend the proposed model by including other relevant constructs under the four key dimensions. Third, the proposed model was tested using one-time cross-sectional data; future research could validate the proposed model using longitudinal data over a period. Fourth, this study attempts to investigate cloud computing adoption in SMEs from owners'/CEOs' perspective; future research may focus employee and cloud provider perspectives to have a broader understanding of the cloud adoption process. Finally, this research examined only organizational intention to adopt cloud technology in SMEs context. The assessment of the successful implementation of this prevalent technology can be a high potential for future research.

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Appendix A. Measurement items

Constructs	Items	Adapted sources
Cloud Adoption	3 items	(Alkhater et al., 2018; Gupta et al., 2013)
Firm performance	4 items	(Garrison et al., 2015; Nagpal and Lyytinen 2010)
Relative advantage	4 items	(Alkhater et al., 2018; Oliveira et al., 2014)
Service Quality	4 items	(Alkhater et al., 2018)
Perceived Risk	4 items	(Johnson et al., 2018; Koenig-Lewis et al., 2015)
Top Management Support	3 items	(Kumar et al., 2017)
Facilitating Conditions	4 items	(Oliveira et al., 2016; Venkatesh et al., 2012)
Cloud Provider's Influence	3 items	(Maqueira-Marin et al., 2017; Repschlaeger et al., 2013)
Server Location	4 items	(Alkhater et al., 2018)
Computer Self Efficacy	3 items	(Ratten 2014; Sharma et al., 2016)
Social Influence	4 items	(Oliveira et al., 2016; Venkatesh et al., 2012)
Resistance to Change	3 items	(Hoque and Sorwar 2017)

Appendix B. Demographic information

Descriptions		Frequency	Percentage
Respondents	Owner/CEO	165	53.1%
	IT manager	146	46.9%
Work experience (Years)	0 to 2	22	7.1%
	3 to 5	54	17.4%
	6 to 10	90	28.9%
	More than 10	145	46.6%
Firm age	Less than 1	8	2.6%
	2 to 5	42	13.5%
	6 to 10	51	16.4%
	11 to 20	89	28.6%

(*continued on next page*)

(*continued*)

Descriptions		Frequency	Percentage
	More than 20	121	38.9%
Firm size (Employees)	Less than 100	92	34.7%
	100 to 200	111	35.7%
	More than 200	108	34.7%
Industry type	Manufacturing	148	47.6%
	Service	163	52.4%
Market scope	Local	55	17.7%
	National	105	33.8%
	International	151	48.6%

Appendix C. Skewness and kurtosis for the research constructs

Statistics	RA	SQ	PR	TMS	FC	CPI	SL	CSE	SI	RC	CA	FP
Skewness	-.547	-.733	-.447	-.096	-.893	-.431	-.640	−1.049	-.695	−1.272	-.816	-.654
Kurtosis	.212	.281	.550	-.574	.871	.275	.267	.646	.242	2.887	.560	.733

Appendix D. Variance inflation factor (VIF)

Constructs	Cloud Adoption	Firm Performance
CA		1.002
FC	2.963	
PR	2.619	
RA	2.479	
RC	1.755	
CSE	1.973	
SI	2.733	
SL	2.577	
SQ	3.105	
TMS	1.478	
CPI	1.984	

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